

## 24.1 Attenuation bias

When a regressor is recorded with error, conditioning on the recorded value is still legitimate, but it describes how the response varies with the *reading*, not with the underlying *quantity*. This section measures the difference.

### 24.1.1 A running example

One simulated study is used throughout the chapter. For  $n = 300$  patients, let  $x_i$  be long-run systolic blood pressure, the average that many visits would give, with mean 130 mmHg and standard deviation 12. Age has mean 55 years, standard deviation 10, and correlation 0.5 with  $x_i$ . Each patient has two clinic readings,  $w_{ij} = x_i + u_{ij}$ , with independent errors  $u_{ij} \sim N(0, 9^2)$ . The outcome is

$$y_i = -4 + 0.08 x_i + 0 \cdot \text{age}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, 1),$$

so age has no effect given long-run pressure. In a real study  $x_i$  is unseen; here it lets every method be checked.

### 24.1.2 Classical error

Throughout the chapter, a regression model is stated in terms of *true* regressors  $\mathbf{x}$  (the ones measured with error) and  $\mathbf{z}$  (the ones measured exactly):

$$Y = \alpha + \boldsymbol{\beta}_x^\top \mathbf{x} + \boldsymbol{\beta}_z^\top \mathbf{z} + \varepsilon, \quad (24.1.1)$$

where  $\varepsilon$  has mean zero and variance  $\sigma^2$  and is uncorrelated with  $(\mathbf{x}, \mathbf{z})$ . We write a generic observation without an index. The data are  $n$  independent copies of it unless stated otherwise.

#### Definition 24.1.1 — Classical measurement error

Suppose  $\mathbf{x}$  is observed only through  $\mathbf{w} = \mathbf{x} + \mathbf{u}$ . The error  $\mathbf{u}$  is **classical** if  $E(\mathbf{u}) = \mathbf{0}$ ,  $\text{Cov}(\mathbf{u}) = \boldsymbol{\Sigma}_{uu}$ , and  $\mathbf{u}$  is independent of  $(\mathbf{x}, \mathbf{z}, \varepsilon)$ .

The error is *additive and unrelated to the true value*, and *nondifferential*: independent of  $\varepsilon$ , so the reading says nothing about the response beyond what  $\mathbf{x}$  says. (People with a disease may recall past exposures differently, a differential error.) This section uses only that  $\mathbf{u}$  has mean zero and is *uncorrelated* with  $\mathbf{x}$ ,  $\mathbf{z}$  and  $\varepsilon$ .

#### Definition 24.1.2 — Reliability ratio

The **reliability ratio** of  $W = X + U$  is

$$\lambda = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_u^2} = \frac{\text{Var}(X)}{\text{Var}(W)}.$$

If  $\mathbf{z}$  is measured exactly, let  $\sigma_{x \cdot z}^2 = \sigma_x^2 - \boldsymbol{\sigma}_{zx}^\top \boldsymbol{\Sigma}_{zz}^{-1} \boldsymbol{\sigma}_{zx}$  be the residual variance of  $X$

after its best linear prediction from  $\mathbf{z}$  (Theorem 6.11.1), where  $\sigma_{zx} = \text{Cov}(\mathbf{z}, X)$ . The **reliability given  $\mathbf{z}$**  is

$$\lambda_{x|z} = \frac{\sigma_{x \cdot z}^2}{\sigma_{x \cdot z}^2 + \sigma_u^2}.$$

In the running example a single reading has  $\lambda = 144/(144 + 81) = 0.64$ . Age explains a quarter of the variance of long-run pressure, leaving  $\sigma_{x \cdot z}^2 = 108$ , so  $\lambda_{x|z} = 108/189 = 0.571$ . Covariates remove part of the signal in  $W$  and none of the noise, so  $\lambda_{x|z} \leq \lambda$ .

### 24.1.3 Least squares on the reading

Substituting  $X = W - U$  in  $Y = \alpha + \beta X + \varepsilon$  gives

$$Y = \alpha + \beta W + (\varepsilon - \beta U). \quad (24.1.2)$$

Its error is correlated with the regressor,  $\text{Cov}(W, \varepsilon - \beta U) = -\beta \sigma_u^2$ , so least squares does not estimate  $\beta$ . It estimates the population projection of  $Y$  on the recorded regressors (Proposition 6.11.3), which the next theorem computes.

#### Theorem 24.1.1 — Attenuation

Let  $(\mathbf{x}_{(i)}, \mathbf{z}_{(i)}, \mathbf{u}_{(i)}, \varepsilon_i)$ ,  $i = 1, 2, \dots$ , be independent copies of  $(\mathbf{x}, \mathbf{z}, \mathbf{u}, \varepsilon)$  with finite second moments, satisfying Eq. 24.1.1 and Definition 24.1.1. Write  $\mathbf{t} = (\mathbf{x}^\top, \mathbf{z}^\top)^\top$  for the true regressors,  $\mathbf{s} = (\mathbf{w}^\top, \mathbf{z}^\top)^\top$  for the observed ones,  $\boldsymbol{\theta} = (\boldsymbol{\beta}_x^\top, \boldsymbol{\beta}_z^\top)^\top$ ,  $\boldsymbol{\Sigma}_{tt} = \text{Cov}(\mathbf{t})$ , assumed positive definite, and  $\tilde{\boldsymbol{\Sigma}} = \text{diag}(\boldsymbol{\Sigma}_{uu}, \mathbf{0})$  for the covariance matrix of the error in  $\mathbf{s}$ . Let  $\hat{\alpha}_n$  and  $\hat{\boldsymbol{\theta}}_n$  be the least squares intercept and slopes from regressing  $Y$  on  $\mathbf{s}$  in the first  $n$  observations. Then, with probability one:

- (a)  $\hat{\boldsymbol{\theta}}_n \rightarrow \boldsymbol{\theta}_w = (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \boldsymbol{\Sigma}_{tt} \boldsymbol{\theta} = \boldsymbol{\theta} - (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta}$ , and  $\hat{\alpha}_n \rightarrow \alpha + \boldsymbol{\mu}_t^\top (\boldsymbol{\theta} - \boldsymbol{\theta}_w)$ , where  $\boldsymbol{\mu}_t = \text{E}(\mathbf{t})$ ;
- (b) with a single regressor and no  $\mathbf{z}$ , the slope converges to  $\lambda\beta$ ;
- (c) with a single mismeasured scalar  $X$  and exactly measured  $\mathbf{z}$ , the coefficient of  $W$  converges to  $\lambda_{x|z}\beta_x$  and the coefficients of  $\mathbf{z}$  converge to  $\boldsymbol{\beta}_z + (1 - \lambda_{x|z})\beta_x\boldsymbol{\pi}$ , where  $\boldsymbol{\pi} = \boldsymbol{\Sigma}_{zz}^{-1}\sigma_{zx}$  is the vector of coefficients from the best linear prediction of  $X$  from  $\mathbf{z}$  (Theorem 2.5.2);
- (d) the residual sum of squares divided by  $n$  (or by  $n$  minus the number of coefficients) converges to  $\sigma^2 + \boldsymbol{\theta}^\top \boldsymbol{\Sigma}_{tt} (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta}$ , which in case (b) equals  $\sigma^2 + \lambda\beta^2\sigma_u^2$ .

*Proof.* Because  $\mathbf{u}$  is uncorrelated with  $\mathbf{t}$ ,  $\text{Cov}(\mathbf{s}) = \boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}}$ , which is positive definite as the sum of a positive definite and a nonnegative definite matrix. By Proposition 6.11.3, the least squares coefficients converge with probability one to those of the best linear predictor of  $Y$  from  $\mathbf{s}$ , which by Theorem 6.11.1 has slopes  $\text{Cov}(\mathbf{s})^{-1} \text{Cov}(\mathbf{s}, Y)$  and intercept  $\text{E}Y - \boldsymbol{\theta}_w^\top \text{E}\mathbf{s}$ . Now

$$\text{Cov}(\mathbf{s}, Y) = \text{Cov}(\mathbf{t} + (\mathbf{u}^\top, \mathbf{0}^\top)^\top, \mathbf{t}^\top \boldsymbol{\theta} + \varepsilon) = \boldsymbol{\Sigma}_{tt} \boldsymbol{\theta},$$

since  $\mathbf{u}$  is uncorrelated with  $\mathbf{t}$  and  $\varepsilon$ , and  $\varepsilon$  with  $\mathbf{t}$ . This is the first form in (a). The second follows from  $\boldsymbol{\Sigma}_{tt} = (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}}) - \tilde{\boldsymbol{\Sigma}}$ . For the intercept,  $\text{E}\mathbf{s} = \boldsymbol{\mu}_t$  and  $\text{E}Y = \alpha + \boldsymbol{\theta}^\top \boldsymbol{\mu}_t$ .

(b) With scalars,  $\theta_w = \sigma_x^2 \beta / (\sigma_x^2 + \sigma_u^2) = \lambda \beta$ .

(c) Here  $\tilde{\Sigma} \theta = \sigma_u^2 \beta_x \mathbf{e}_1$ , so by the second form in (a) the bias  $\theta - \theta_w$  is  $\sigma_u^2 \beta_x$  times the first column of  $\text{Cov}(\mathbf{s})^{-1}$ . Order the observed regressors as  $(\mathbf{z}, W)$  for the moment, so that

$$\text{Cov}(\mathbf{s}) = \begin{pmatrix} \Sigma_{zz} & \sigma_{zx} \\ \sigma_{zx}^\top & \sigma_x^2 + \sigma_u^2 \end{pmatrix}.$$

The Schur complement of  $\Sigma_{zz}$  is  $\sigma_x^2 + \sigma_u^2 - \sigma_{zx}^\top \Sigma_{zz}^{-1} \sigma_{zx} = \sigma_{x \cdot z}^2 + \sigma_u^2$ . By Theorem 1.6.4 the last column of the inverse is  $(-\pi^\top, 1)^\top / (\sigma_{x \cdot z}^2 + \sigma_u^2)$ . So the bias in the coefficient of  $W$  is  $\sigma_u^2 \beta_x / (\sigma_{x \cdot z}^2 + \sigma_u^2) = (1 - \lambda_{x|z}) \beta_x$ , and the bias in the coefficients of  $\mathbf{z}$  is  $-(1 - \lambda_{x|z}) \beta_x \pi$ .

(d) The residual sum of squares over  $n$  is  $n^{-1} \sum_i (Y_i - \hat{\alpha}_n - \hat{\theta}_n^\top \mathbf{s}_i)^2$ . Expanded, it is a continuous function of the coefficients and of second sample moments, which converge by (a) and the strong law, so the limit is  $E(Y - \alpha_w - \theta_w^\top \mathbf{s})^2$ , the mean squared error of the best linear predictor. By Eq. 6.11.1 it equals  $\text{Var}(Y) - \theta_w^\top \text{Cov}(\mathbf{s}) \theta_w$ , which is

$$\sigma^2 + \theta^\top \Sigma_{tt} \theta - \theta^\top \Sigma_{tt} (\Sigma_{tt} + \tilde{\Sigma})^{-1} \Sigma_{tt} \theta = \sigma^2 + \theta^\top \Sigma_{tt} (\Sigma_{tt} + \tilde{\Sigma})^{-1} \tilde{\Sigma} \theta.$$

In case (b) the last term is  $\beta^2 \sigma_x^2 \sigma_u^2 / (\sigma_x^2 + \sigma_u^2) = \lambda \beta^2 \sigma_u^2$ .

■

Part (b) is **attenuation**: the slope is shrunk towards zero by the reliability ratio, whatever the sample size. More data give a more precise estimate of the wrong number  $\lambda \beta$ .

#### Example. (One reading instead of long-run pressure)

In the running example, regressing  $y$  on the true  $x$  gives slope 0.0755 (standard error 0.0050). Regressing on the first clinic reading gives 0.0499 (standard error 0.0046), a ratio of 0.661, close to  $\lambda = 0.64$ ; the limit is  $\lambda \beta = 0.0512$ . The usual 95% interval, [0.0409, 0.0589], misses 0.08, and in 4000 repeated samples the observed coverage was 0.000. The standard error measures variability around  $\lambda \beta$ , not the distance from  $\lambda \beta$  to  $\beta$ . The residual variance tends to  $1 + 0.64 \cdot 0.08^2 \cdot 81 = 1.3318$  by Theorem 24.1.1(d).

### 24.1.4 Several regressors: bias spreads

Part (c) is more troubling. Correctly measured covariates lower the relevant reliability to  $\lambda_{x|z}$ , and they inherit part of the effect of  $X$ . The part of  $X$  that the reading misses acts like an omitted regressor (Theorem 19.1.2), and covariates correlated with  $X$  pick up its effect in proportion to  $\pi$ . This is **residual confounding**: adjusting for a noisy measurement of a confounder only partly adjusts for it.

#### Example. (Age looks important)

Add age to the regression on the first reading. Theory predicts a coefficient of  $\lambda_{x|z} \beta = 0.0457$  for the reading and  $(1 - 0.571) \cdot 0.08 \cdot 0.6 = 0.0206$  per year for age, which truly has no effect. The sample gives 0.0455 for the reading and 0.0182 for age, with standard error 0.0070 and  $p$ -value 0.0098. In 4000 repeated samples the  $t$  test of “no age effect” rejected at the 5% level in a proportion 0.812 of them. Figure 24.1.1(b) shows both

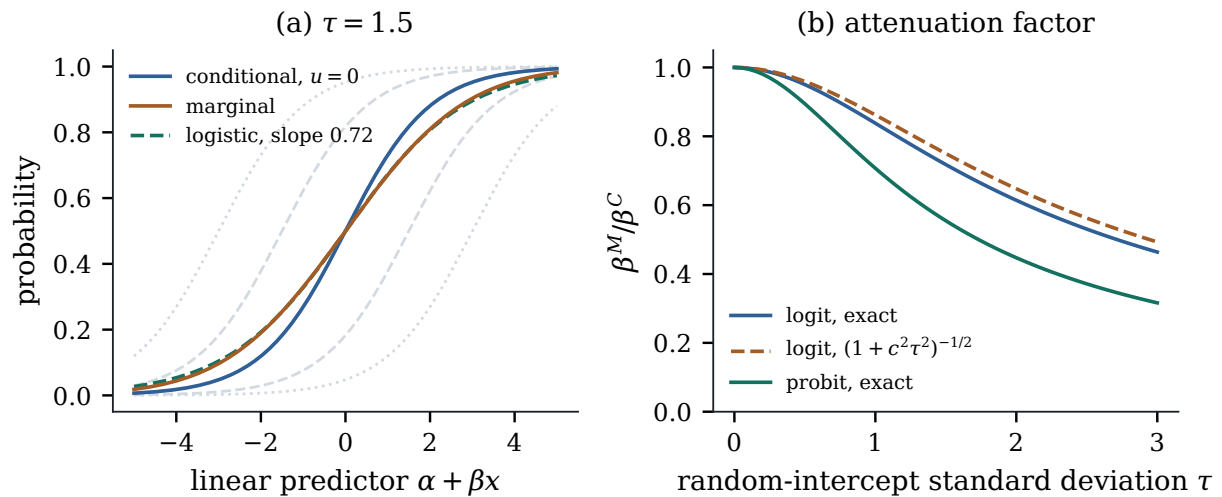


Figure 24.1.1: (a) The outcomes against true pressure (grey) and one reading (blue), with their least squares lines. (b) Limits of the coefficients of the reading and of age, in units of  $\beta_x$ , against the reliability; dashed,  $\lambda$  itself.

limits as functions of the reliability.

```
import numpy as np
import statsmodels.api as sm

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                        pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2))  # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n)  # age has no effect
                        given x

oracle = sm.OLS(y, sm.add_constant(x)).fit()  # uses the true x
naive = sm.OLS(y, sm.add_constant(w[:, 0])).fit()  # uses one reading
lam = 144 / (144 + 81)  # reliability of one
                        reading

print(f"slope on x: {oracle.params[1]:.4f}")
print(f"slope on reading: {naive.params[1]:.4f} limit lambda*beta = {lam *
0.08:.4f}")
print("naive 95% interval:", naive.conf_int()[1].round(4))

both = sm.OLS(y, sm.add_constant(np.column_stack([w[:, 0], age]))).fit()
print(f"with age: reading {both.params[1]:.4f}, age {both.params[2]:.4f} (p = {both.
pvalues[2]:.4f})")
```

[Run this code online](#)

The repeated-sample figures come from this loop (with fewer samples in the cell).

```
def repeat(reps, seed):
    """Coverage of the naive 95% interval for the slope, and the rate at which age
    is 'significant'."""
    r = np.random.default_rng(seed)
    cover = reject = 0
```

```

for _ in range(reps):
    a = r.normal(55, 10, n)
    xx = 130 + 0.6 * (a - 55) + r.normal(0, np.sqrt(108), n)
    ww = xx + r.normal(0, 9, n)
    yy = -4 + 0.08 * xx + r.normal(0, 1.0, n)
    lo, hi = sm.OLS(yy, sm.add_constant(ww)).fit().conf_int()[1]
    cover += lo <= 0.08 <= hi
    reject += sm.OLS(yy, sm.add_constant(np.column_stack([ww, a]))).fit().
pvalues[2] < 0.05
return cover / reps, reject / reps

print("coverage of naive interval, rate of rejecting 'no age effect':", repeat(200,
2404))

```

[Run this code online](#)

With several mismeasured regressors,  $(\Sigma_{tt} + \tilde{\Sigma})^{-1}\Sigma_{tt}$  need not shrink each coefficient. A coefficient can grow, change sign, or appear from nothing ([Exercise \(B3\)](#)).

## 24.1.5 What survives

Two things survive classical error intact.

**Error in the response.** Suppose instead that the response is measured with error,  $Y^* = Y + V$ , where  $V$  has mean zero and variance  $\sigma_v^2$  and is independent of the regressors and of  $\varepsilon$ .

### Proposition 24.1.2 — Error in the response

Under these assumptions  $Y^* = \alpha + \beta^\top \mathbf{x} + (\varepsilon + V)$ , a linear model with the same coefficients and error variance  $\sigma^2 + \sigma_v^2$ . If  $\varepsilon$  and  $V$  are normal, so is their sum, and all the exact theory of Parts II and III applies to  $Y^*$  with  $\sigma^2$  replaced by  $\sigma^2 + \sigma_v^2$ .

The proof is the display. Error in the response costs precision, not bias, unless it depends on the regressors (heavier people under-reporting weight) or is correlated with a regressor's error (both from the same self-report).

**Tests of no effect.** Since  $\lambda\beta = 0$  iff  $\beta = 0$ , attenuation can hide an effect of the mismeasured regressor but cannot create one. The protection does not extend to the covariates.

### Proposition 24.1.3 — Which tests survive classical error

Assume [Eq. 24.1.1](#) and [Definition 24.1.1](#) with a single mismeasured scalar  $X$ ,  $\varepsilon \sim N(0, \sigma^2)$  independent of  $(X, \mathbf{z})$ , and the covariance matrix of  $(X, \mathbf{z}^\top)^\top$  positive definite. Consider the least squares regression of  $Y$  on  $(1, W, \mathbf{z})$ .

- (a) If  $\beta_x = 0$ , the usual  $t$  test of “the coefficient of  $W$  is zero” has exactly its nominal level.
- (b) If  $\beta_x \neq 0$  and  $\boldsymbol{\pi} \neq \mathbf{0}$ , the usual  $F$  test of “the coefficients of  $\mathbf{z}$  equal  $\boldsymbol{\beta}_z$ ” rejects with probability tending to one as  $n \rightarrow \infty$ .

*Proof.* (a) If  $\beta_x = 0$ , then  $Y = \alpha + \beta_z^\top \mathbf{z} + \varepsilon$ , and  $\varepsilon$  is independent of  $(W, \mathbf{z})$  because it is independent of  $(X, \mathbf{z}, U)$ : it is independent of  $(X, \mathbf{z})$  by hypothesis, and  $U$  is independent of  $(X, \mathbf{z}, \varepsilon)$  by Definition 24.1.1. So, conditionally on the observed regressors,  $Y$  follows the normal linear model of Definition 14.1.1 with regressors  $(1, W, \mathbf{z})$ , in which the coefficient of  $W$  is zero. Theorem 14.1.2 gives the exact  $t$  distribution.

(b) By Theorem 24.1.1(c),  $\hat{\beta}_z - \beta_z \rightarrow \mathbf{c} = (1 - \lambda_{x|z})\beta_x \boldsymbol{\pi} \neq \mathbf{0}$ . The  $F$  statistic is  $n(\hat{\beta}_z - \beta_z)^\top \mathbf{K}_n(\hat{\beta}_z - \beta_z)$ , where  $\mathbf{K}_n = [n(\mathbf{S}^\top \mathbf{S})^{-1}]_{zz}^{-1}/(q s^2)$ ,  $\mathbf{S}$  is the observed model matrix and  $q$  the dimension of  $\mathbf{z}$ . By the strong law and Theorem 24.1.1(d),  $\mathbf{K}_n$  converges to a positive definite limit, so the statistic grows like  $n$ , while the critical value tends to a finite limit.

■

So “is there any association with the underlying quantity?” can be tested with the reading, with reduced power. “Does this covariate matter once the underlying quantity is accounted for?” cannot.

## 24.1.6 Exercises

### 24.1.6.1 A. Check your understanding

#### Exercise A1.

Show that  $\lambda = \text{Corr}(W, X)^2$ . If  $W_1 = X + U_1$  and  $W_2 = X + U_2$  are two readings with independent classical errors of the same variance, show that  $\text{Corr}(W_1, W_2) = \lambda$ . This is why test–retest correlations are used as reliability estimates.

*Solution.*  $\text{Cov}(W, X) = \sigma_x^2$ , so  $\text{Corr}(W, X)^2 = \sigma_x^4 / \{\sigma_x^2(\sigma_x^2 + \sigma_u^2)\} = \lambda$ . For two readings,  $\text{Cov}(W_1, W_2) = \text{Var}(X) = \sigma_x^2$  because the errors are independent of each other and of  $X$ , and  $\text{Var}(W_1) = \text{Var}(W_2) = \sigma_x^2 + \sigma_u^2$ . The correlation is  $\sigma_x^2 / (\sigma_x^2 + \sigma_u^2) = \lambda$ .

### 24.1.6.2 B. Practice

#### Exercise B1.

Let  $\bar{W}$  be the mean of  $m$  readings with independent classical errors of variance  $\sigma_u^2$ , and let  $\lambda$  be the reliability of one reading. Show that the reliability of  $\bar{W}$  is

$$\lambda_m = \frac{m\lambda}{1 + (m-1)\lambda}.$$

How many readings are needed in the running example for the attenuation factor to reach 0.9?

*Solution.*  $\lambda_m = \sigma_x^2 / (\sigma_x^2 + \sigma_u^2/m)$ . Dividing through by  $\sigma_x^2 + \sigma_u^2$  and using  $\sigma_u^2 / (\sigma_x^2 + \sigma_u^2) = 1 - \lambda$  gives  $\lambda / \{\lambda + (1 - \lambda)/m\}$ , the formula. With  $\lambda = 0.64$ ,  $\lambda_m \geq 0.9$  iff  $0.64m \geq 0.9 + 0.576(m - 1)$ , so  $m \geq 5.0625$ : six readings (without age in the model).

**Exercise B2.**

In the setting of [Theorem 24.1.1\(b\)](#), show that the population  $R^2$  of  $Y$  on  $W$  equals  $\lambda$  times the population  $R^2$  of  $Y$  on  $X$ . Measurement error lowers the apparent strength of a relation twice: once through the slope and once through the residual variance.

**Exercise B3.**

Let  $\mathbf{t} = (X_1, X_2)$  have unit variances and correlation  $\rho$ , let  $X_1$  be measured exactly and  $X_2$  with classical error of variance  $\sigma_u^2$ , and let  $\boldsymbol{\theta} = (\beta_1, \beta_2)$ . Show that the limit of the coefficient of  $X_1$  is  $\beta_1 + (1 - \lambda_{2|1})\rho\beta_2$ , where  $\lambda_{2|1} = (1 - \rho^2)/(1 - \rho^2 + \sigma_u^2)$ . With  $\rho = 0.5$ ,  $\sigma_u^2 = 1$ ,  $\beta_2 = 1$  and  $\beta_1 = -0.2$ , show that the sign of the first coefficient is reversed.

*Solution.* This is [Theorem 24.1.1\(c\)](#) with  $\mathbf{z} = X_1$ ,  $X = X_2$  and  $\boldsymbol{\pi} = \rho$ , since the coefficient of the best linear prediction of  $X_2$  from  $X_1$  is  $\rho$  and  $\sigma_{x \cdot z}^2 = 1 - \rho^2$ . With the numbers given,  $\lambda_{2|1} = 0.75/1.75 = 3/7$ , so the limit is  $-0.2 + (4/7)(0.5) = -0.2 + 2/7 \approx 0.086 > 0$ . A regressor with a negative effect acquires a positive coefficient.

**24.1.6.3 C. Going deeper****Exercise C1.**

Regressors are often recorded after rounding. Model this as  $\mathbf{S} = \mathbf{T} + \boldsymbol{\Delta}$  with  $\mathbf{T}$  the true model matrix and  $\boldsymbol{\Delta}$  a *fixed* matrix of rounding errors, and suppose  $E(\mathbf{Y}) = \mathbf{T}\boldsymbol{\theta}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ . Use Proposition 5.5.7 to show that least squares on  $\mathbf{S}$  has bias  $-(\mathbf{S}^\top\mathbf{S})^{-1}\mathbf{S}^\top\boldsymbol{\Delta}\boldsymbol{\theta}$ , that its covariance matrix is exactly  $\sigma^2(\mathbf{S}^\top\mathbf{S})^{-1}$ , and that the expected residual mean square is at least  $\sigma^2$ .

*Solution.* Here  $E(\mathbf{Y}) = \mathbf{S}\boldsymbol{\theta} + \boldsymbol{\delta}$  with  $\boldsymbol{\delta} = -\boldsymbol{\Delta}\boldsymbol{\theta}$ . By Proposition 5.5.7(a) with model matrix  $\mathbf{S}$ , the expectation is  $\boldsymbol{\theta} + (\mathbf{S}^\top\mathbf{S})^{-1}\mathbf{S}^\top\boldsymbol{\delta}$ , which gives the bias. The covariance is  $\sigma^2(\mathbf{S}^\top\mathbf{S})^{-1}$  by Proposition 5.5.7(b) with  $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ : nothing random has been added. For the residual mean square, Theorem 2.3.1 gives  $E\|(\mathbf{I} - \mathbf{M}_S)\mathbf{Y}\|^2 = \sigma^2(n-p) + \boldsymbol{\delta}^\top(\mathbf{I} - \mathbf{M}_S)\boldsymbol{\delta}$ , where  $\mathbf{M}_S$  projects onto  $\mathcal{C}(\mathbf{S})$  and  $(\mathbf{I} - \mathbf{M}_S)\mathbf{S}\boldsymbol{\theta} = \mathbf{0}$ . The last term is nonnegative. Unlike random error, fixed rounding error does not inflate the variance of the estimator. It only biases it, and the bias is small when rounding is fine compared with the spread of the regressors.